Murilo Corato Zanarella murilocorato@gmail.com • murilocorato.github.io

employment

1. Johns Hopkins University, J.J. Sylvester Assistant Professor 2024-2026

education

2. Massachusetts Institute of Technology, PhD in Mathematics 2019-2024
   Advisor: Wei Zhang
1. Princeton University, B.A. Mathematics 2015-2019

research interests

I am interested in arithmetic aspects of the (relative) Langlands program, particularly in the role of Shimura varieties, and applications to the Beilinson–Bloch–Kato conjectures, Euler systems and Iwasawa theory.

papers

8. Relative Satake transforms and arithmetic level raising for unitary Friedberg–Jacquet periods 2026
   in preparation
7. Potential maps beyond the strictly semistable case 2026
   in preparation
6. Unequal-parameter Kostka–Foulkes polynomials of type C_n at fundamental weights 2026
   preprint, `https://arxiv.org/abs/2609.39095`
5. First explicit reciprocity law for unitary Friedberg-Jacquet periods 2026
   submitted, `https://arxiv.org/abs/2602.09831`
4. Spherical functions on symmetric spaces of Friedberg-Jacquet type 2023
   submitted, `https://arxiv.org/abs/2311.00148`
3. Spherical functions of symmetric forms and a conjecture of Hironaka 2023
   submitted, `https://arxiv.org/abs/2311.00147`
2. On Howard's main conjecture and the Heegner point Kolyvagin system 2019
   Undergraduate senior thesis, preprint, `https://arxiv.org/abs/1908.09197`
1. A proof of Kolyvagin's conjecture via the BDP main conjecture 2019
   Undergraduate junior paper, preprint, `https://arxiv.org/abs/1909.07835`

invited talks

11. MIT: Bipartite Euler systems and local harmonic analysis November 2026
10. Nisyros: Bipartite Euler systems and local harmonic analysis July 2026
9. ANTD: Bipartite Euler systems and local harmonic analysis March 2026
8. Columbia: 1st reciprocity law for unitary FJ periods February 2025
7. Duke: 1st reciprocity law for unitary FJ periods October 2024
6. IAS: 1st reciprocity law for unitary FJ periods October 2024
5. UT Austin: 1st reciprocity law for unitary FJ periods February 2024
4. JHU/JNTD: 1st reciprocity law for unitary FJ periods February 2024
3. TSIMF: 1st reciprocity law for unitary FJ periods January 2024
2. MSRI/SLMath: Arithmetic level raising and reciprocity laws Mar 2023
1. JMM: On Howard's MC and the Heegner point Kolyvagin system Jan 2020

organizing

3. Number theory seminar (JHU, co-organizer) — Fall 2025 - Spring 2027
2. Number theory learning seminar (MIT, co-organizer) — Fall 2024
1. Learning seminar on Euler systems (MIT) — Fall 2022

teaching

14. 110.617 Number Theory I — Fall 2026
13. 110.109 Calculus II (for Physical Sciences and Engineering) — Spring 2026
12. 110.422 Representation Theory — Fall 2025
11. 110.617 Number Theory I — Fall 2025
10. 110.202 Calculus III — Spring 2025
9. 110.617 Number Theory I — Fall 2024
8. 18.701 Algebra I (Teaching Assistant) — Fall 2023
7. 18.950 Differential Geometry (Teaching Assistant) — Fall 2023
6. 18.02 Multivariable Calculus (Teaching Assistant) — Fall 2022
5. 18.065 Matrix Methods in Data Analysis & ML (TA) — Spring 2021
4. 18.701 Algebra I (Teaching Assistant) — Fall 2021
3. 18.700 Linear Algebra (Teaching Assistant) — Fall 2021
2. 18.702 Algebra II (Teaching Assistant) — Spring 2020
1. 18.100A Real Analysis (Teaching Assistant) — Fall 2020

mentorship

5. JHU reading course — Spring 2026
   Mentored an undergraduate on a first course in modular forms
4. JHU reading course — Spring 2025
   Mentored an undergraduate on the arithmetic of elliptic curves
3. High School Enrichment Program Teacher — 2021-2024
   Virtual classes with students from my former high school on undergraduate-level topics in number theory
2. MIT Directed Reading Program — Winter 2020
   Mentored 2 undergraduates on analytic number theory
1. MIT Directed Reading Program — Winter 2020
   Mentored 3 undergraduates on modular forms and elliptic curves

awards

8. Professor Joel Dean Award for Excellence in Teaching — 2025
   Awarded for extraordinary performance in teaching undergraduates.
7. Frank and Brennie Morgan Prize (hon. mention), AMS/MAA/SIAM — 2020
   Awarded for outstanding research in mathematics by an undergraduate
6. The Middleton Miller '29 prize — 2018
   Awarded for the best independent work in mathematics
5. Peter A. Greenberg '77 Memorial Prize — 2018
   Awarded for outstanding accomplishments in mathematics
4. Putnam examination — 2016-2018
   N1 prize (6th-14th) in 2016 and 2018, Honorable mention in 2017
3. Shapiro prize for academic excellence — 2017, 2018
   Award for outstanding academic achievement
2. The Class of 1861 prize — 2017
   Awarded to the sophomore with the best record on the Putnam
1. International mathematics olympiad — 2014-2015
   Silver medals in 2014 and 2015